$$
\begin{aligned}
&C_{\phi\,\partial^2}^{(11)\ (11)} \to \frac{1}{16\pi^2}\left(-\frac{1}{6} g_2^4\, \mathrm{LF}_{3,0}[m_2] - \frac{1}{4} g_2^4\, \mathrm{LF}_{4,-1}[m_2] + \frac{4}{15} g_2^4\, \mathrm{LF}_{5,-2}[m_2] -\right.\\
&\frac{1}{24}\sum_p g_1^4\, \mathrm{LF}_{3,0}[m_{\tilde d}^p] + \frac{1}{24}\sum_p g_1^4\, \mathrm{LF}_{4,-1}[m_{\tilde d}^p] - \frac{1}{8}\sum_p g_1^4\, \mathrm{LF}_{3,0}[m_{\tilde e}^p] + \frac{1}{8}\sum_p g_1^4\, \mathrm{LF}_{4,-1}[m_{\tilde e}^p] +\\
&\frac{1}{48}\sum_p\left(-3g_1^4+2g_2^4\right)\mathrm{LF}_{3,0}[m_{\tilde l}^p] + \frac{1}{32}\sum_p\left(2g_1^4+g_2^4\right)\mathrm{LF}_{4,-1}[m_{\tilde l}^p] -\\
&\frac{1}{15}\sum_p g_2^4\, \mathrm{LF}_{5,-2}[m_{\tilde l}^p] + \left(-\frac{1}{4}\overline{y_u}^{pr} y_u^{pr}\left(g_1^2+6g_2^2\right) - \frac{1}{48}\sum_p\left(g_1^4-6g_2^4\right)\right)\mathrm{LF}_{3,0}[m_{\tilde q}^p] +\\
&\frac{1}{96}\left(24\,\overline{y_u}^{pr} y_u^{pr}\left(g_1^2+6g_2^2\right)+\sum_p\left(2g_1^4+9g_2^4\right)\right)\mathrm{LF}_{4,-1}[m_{\tilde q}^p] - \frac{1}{5}\sum_p g_2^4\, \mathrm{LF}_{5,-2}[m_{\tilde q}^p] -\\
&\frac{1}{6}\sum_p g_1^4\, \mathrm{LF}_{3,0}[m_{\tilde u}^p] + \frac{1}{6}\sum_p g_1^4\, \mathrm{LF}_{4,-1}[m_{\tilde u}^p] + g_1^2\,\overline{y_u}^{pr} y_u^{pr}\, \mathrm{LF}_{3,0}[m_{\tilde u}^r] -\\
&g_1^2\,\overline{y_u}^{pr} y_u^{pr}\, \mathrm{LF}_{4,-1}[m_{\tilde u}^r] - \frac{1}{12} g_2^4\, \mathrm{LF}_{3,0}[\tilde\mu] - \frac{1}{8} g_2^4\, \mathrm{LF}_{4,-1}[\tilde\mu] + \frac{2}{15} g_2^4\, \mathrm{LF}_{5,-2}[\tilde\mu] -\\
&\frac{7}{32} g_1^4\, \mathrm{LF}_{2,2,-1}[m_1,\tilde\mu] - \frac{1}{16} g_1^4 m_1^2\, \mathrm{LF}_{2,2,0}[m_1,\tilde\mu] + \frac{1}{2} g_1^4\, \mathrm{LF}_{3,2,-2}[m_1,\tilde\mu] +\\
&\frac{3}{8} g_1^4 m_1^2\, \mathrm{LF}_{3,2,-1}[m_1,\tilde\mu] - \frac{1}{4} g_1^4\, \mathrm{LF}_{3,3,-3}[m_1,\tilde\mu] - \frac{1}{4} g_1^4 m_1^2\, \mathrm{LF}_{3,3,-2}[m_1,\tilde\mu] -\\
&\frac{1}{4} g_1^4\, \mathrm{LF}_{4,2,-3}[m_1,\tilde\mu] - \frac{1}{4} g_1^4 m_1^2\, \mathrm{LF}_{4,2,-2}[m_1,\tilde\mu] + g_2^4\, \mathrm{LF}_{2,1,0}[m_2,\tilde\mu] + \frac{33}{32} g_2^4\, \mathrm{LF}_{2,2,-1}[m_2,\tilde\mu] -\\
&\frac{1}{16} g_2^4 m_2^2\, \mathrm{LF}_{2,2,0}[m_2,\tilde\mu] - \frac{1}{2} g_2^4\, \mathrm{LF}_{3,1,-1}[m_2,\tilde\mu] - g_2^4\, \mathrm{LF}_{3,2,-2}[m_2,\tilde\mu] +\\
&\frac{21}{8} g_2^4 m_2^2\, \mathrm{LF}_{3,2,-1}[m_2,\tilde\mu] + \frac{1}{4} g_2^4\, \mathrm{LF}_{3,3,-3}[m_2,\tilde\mu] - \frac{7}{4} g_2^4 m_2^2\, \mathrm{LF}_{3,3,-2}[m_2,\tilde\mu] +\\
&\frac{1}{4} g_2^4\, \mathrm{LF}_{4,2,-3}[m_2,\tilde\mu] - \frac{7}{4} g_2^4 m_2^2\, \mathrm{LF}_{4,2,-2}[m_2,\tilde\mu] - \frac{1}{2} g_1^2\tilde\mu^2\,\overline{y_d}^{pr} y_d^{pr}\, \mathrm{LF}_{3,1,0}[m_{\tilde d}^r, m_{\tilde q}^p] +\\
&\frac{1}{4} g_1^2\tilde\mu^2\,\overline{y_d}^{pr} y_d^{pr}\, \mathrm{LF}_{3,2,-1}[m_{\tilde d}^r, m_{\tilde q}^p] + \frac{1}{2} g_1^2\tilde\mu^2\,\overline{y_d}^{pr} y_d^{pr}\, \mathrm{LF}_{4,1,-1}[m_{\tilde d}^r, m_{\tilde q}^p] -\\
&\frac{1}{2} g_1^2\tilde\mu^2\,\overline{y_e}^{pr} y_e^{pr}\, \mathrm{LF}_{3,1,0}[m_{\tilde e}^r, m_{\tilde l}^p] + \frac{1}{4} g_1^2\tilde\mu^2\,\overline{y_e}^{pr} y_e^{pr}\, \mathrm{LF}_{3,2,-1}[m_{\tilde e}^r, m_{\tilde l}^p] +\\
&\frac{1}{2} g_1^2\tilde\mu^2\,\overline{y_e}^{pr} y_e^{pr}\, \mathrm{LF}_{4,1,-1}[m_{\tilde e}^r, m_{\tilde l}^p] + \frac{1}{8}\tilde\mu^2\,\overline{y_e}^{pr} y_e^{pr}\left(g_1^2-g_2^2\right)\mathrm{LF}_{4,1,-1}[m_{\tilde l}^p, m_{\tilde e}^r] +\\
&\frac{1}{2} g_2^2\tilde\mu^2\,\overline{y_e}^{pr} y_e^{pr}\, \mathrm{LF}_{5,1,-2}[m_{\tilde l}^p, m_{\tilde e}^r] - \frac{1}{8}\tilde\mu^2\,\overline{y_d}^{pr} y_d^{pr}\left(g_1^2+3g_2^2\right)\mathrm{LF}_{4,1,-1}[m_{\tilde q}^p, m_{\tilde d}^r] +\\
&\frac{3}{2} g_2^2\tilde\mu^2\,\overline{y_d}^{pr} y_d^{pr}\, \mathrm{LF}_{5,1,-2}[m_{\tilde q}^p, m_{\tilde d}^r] + \frac{3}{4}\overline{y_u}^{pr}\,\overline{y_u}^{st} y_u^{pt} y_u^{sr}\, \mathrm{LF}_{2,1,0}[m_{\tilde q}^p, m_{\tilde q}^s] -\\
&\frac{3}{4}\overline{y_u}^{pr}\,\overline{y_u}^{st} y_u^{pt} y_u^{sr}\, \mathrm{LF}_{3,1,-1}[m_{\tilde q}^p, m_{\tilde q}^s] - \frac{3}{4} g_2^2\,\overline{a_u}^{pr} a_u^{pr}\, \mathrm{LF}_{2,2,0}[m_{\tilde q}^p, m_{\tilde u}^r] -\\
&\frac{1}{4}\overline{a_u}^{pr} a_u^{pr}\left(g_1^2+6g_2^2\right)\mathrm{LF}_{3,1,0}[m_{\tilde q}^p, m_{\tilde u}^r] + \frac{1}{8}\overline{a_u}^{pr} a_u^{pr}\left(g_1^2+9g_2^2\right)\mathrm{LF}_{3,2,-1}[m_{\tilde q}^p, m_{\tilde u}^r] +\\
&\frac{1}{4}\overline{a_u}^{pr} a_u^{pr}\left(g_1^2+6g_2^2\right)\mathrm{LF}_{4,1,-1}[m_{\tilde q}^p, m_{\tilde u}^r] + \frac{3}{4} g_2^2\,\overline{a_u}^{pr} a_u^{pr}\, \mathrm{LF}_{3,1,0}[m_{\tilde u}^r, m_{\tilde q}^p] +\\
&\frac{3}{4} g_2^2\,\overline{a_u}^{pr} a_u^{pr}\, \mathrm{LF}_{3,2,-1}[m_{\tilde u}^r, m_{\tilde q}^p] + \frac{1}{4}\overline{a_u}^{pr} a_u^{pr}\left(2g_1^2-9g_2^2\right)\mathrm{LF}_{4,1,-1}[m_{\tilde u}^r, m_{\tilde q}^p] +\\
&\frac{3}{2} g_2^2\,\overline{a_u}^{pr} a_u^{pr}\, \mathrm{LF}_{5,1,-2}[m_{\tilde u}^r, m_{\tilde q}^p] - \frac{3}{2}\overline{y_u}^{pr}\,\overline{y_u}^{st} y_u^{pt} y_u^{sr}\, \mathrm{LF}_{2,1,0}[m_{\tilde u}^r, m_{\tilde u}^t] +\\
&\frac{3}{2}\overline{y_u}^{pr}\,\overline{y_u}^{st} y_u^{pt} y_u^{sr}\, \mathrm{LF}_{3,1,-1}[m_{\tilde u}^r, m_{\tilde u}^t] + \frac{3}{16} g_1^4\, \mathrm{LF}_{3,2,-2}[\tilde\mu, m_1] + \frac{1}{16} g_1^4 m_1^2\, \mathrm{LF}_{3,2,-1}[\tilde\mu, m_1] +\\
&\frac{5}{8} g_1^2 g_2^2\, \mathrm{LF}_{4,1,-2}[\tilde\mu, m_1] - \frac{1}{16} g_1^4\, \mathrm{LF}_{4,2,-3}[\tilde\mu, m_1] - \frac{1}{16} g_1^4 m_1^2\, \mathrm{LF}_{4,2,-2}[\tilde\mu, m_1] -\\
&\frac{1}{2} g_1^2 g_2^2\, \mathrm{LF}_{5,1,-3}[\tilde\mu, m_1] - 2 g_2^4\, \mathrm{LF}_{3,1,-1}[\tilde\mu, m_2] - \frac{25}{16} g_2^4\, \mathrm{LF}_{3,2,-2}[\tilde\mu, m_2] +\\
&\frac{1}{16} g_2^4 m_2^2\, \mathrm{LF}_{3,2,-1}[\tilde\mu, m_2] + \frac{27}{8} g_2^4\, \mathrm{LF}_{4,1,-2}[\tilde\mu, m_2] + \frac{7}{16} g_2^4\, \mathrm{LF}_{4,2,-3}[\tilde\mu, m_2] -\\
&\frac{1}{16} g_2^4 m_2^2\, \mathrm{LF}_{4,2,-2}[\tilde\mu, m_2] - \frac{3}{2} g_2^4\, \mathrm{LF}_{5,1,-3}[\tilde\mu, m_2] - 2 g_1^2 g_2^2\, \mathrm{LF}_{2,2,1,-2}[m_2,\tilde\mu, m_1] -\\
&\frac{3}{2} m_1 m_2 g_1^2 g_2^2\, \mathrm{LF}_{2,2,1,-1}[m_2,\tilde\mu, m_1] + g_1^2 g_2^2\, \mathrm{LF}_{3,2,1,-3}[m_2,\tilde\mu, m_1] +\\
&m_1 m_2 g_1^2 g_2^2\, \mathrm{LF}_{3,2,1,-2}[m_2,\tilde\mu, m_1] - \frac{9}{4}\tilde\mu^2\,\overline{y_d}^{pr} y_d^{sr}\,\overline{y_u}^{st} y_u^{pt}\, \mathrm{LF}_{2,2,1,-1}[m_{\tilde q}^p, m_{\tilde q}^s, m_{\tilde d}^r] +\\
&\frac{3}{4}\overline{y_u}^{st} y_u^{pt}\,\overline{a_u}^{pr} a_u^{sr}\, \mathrm{LF}_{2,1,1,0}[m_{\tilde q}^p, m_{\tilde q}^s, m_{\tilde u}^r] - \frac{3}{4}\overline{y_u}^{st} y_u^{pt}\,\overline{a_u}^{pr} a_u^{sr}\, \mathrm{LF}_{3,1,1,-1}[m_{\tilde q}^p, m_{\tilde q}^s, m_{\tilde u}^r] -\\
&\frac{3}{8}\overline{y_u}^{st} y_u^{pt}\,\overline{a_u}^{pr} a_u^{sr}\, \mathrm{LF}_{2,2,1,-1}[m_{\tilde q}^p, m_{\tilde u}^r, m_{\tilde q}^s] + \frac{3}{4}\overline{y_u}^{st} y_u^{pt}\,\overline{a_u}^{pr} a_u^{sr}\, \mathrm{LF}_{2,1,1,0}[m_{\tilde q}^s, m_{\tilde q}^p, m_{\tilde u}^r] -\\
&\frac{3}{4}\overline{y_u}^{st} y_u^{pt}\,\overline{a_u}^{pr} a_u^{sr}\, \mathrm{LF}_{3,1,1,-1}[m_{\tilde q}^s, m_{\tilde q}^p, m_{\tilde u}^r] - \frac{3}{8}\overline{y_u}^{st} y_u^{pt}\,\overline{a_u}^{pr} a_u^{sr}\, \mathrm{LF}_{2,2,1,-1}[m_{\tilde q}^s, m_{\tilde u}^r, m_{\tilde q}^p] -\\
&\frac{3}{2}\overline{y_u}^{st} y_u^{sr}\,\overline{a_u}^{pr} a_u^{pt}\, \mathrm{LF}_{2,2,1,-1}[m_{\tilde u}^r, m_{\tilde u}^t, m_{\tilde q}^p] + \frac{11}{16} g_1^2 g_2^2\, \mathrm{LF}_{2,1,1,-1}[\tilde\mu, m_1, m_2] -\\
&\frac{1}{8} m_1 m_2 g_1^2 g_2^2\, \mathrm{LF}_{2,1,1,0}[\tilde\mu, m_1, m_2] - \frac{9}{8} g_1^2 g_2^2\, \mathrm{LF}_{3,1,1,-2}[\tilde\mu, m_1, m_2] +\\
&\frac{1}{8} m_1 m_2 g_1^2 g_2^2\, \mathrm{LF}_{3,1,1,-1}[\tilde\mu, m_1, m_2] + \frac{5}{8} g_1^2 g_2^2\, \mathrm{LF}_{4,1,1,-3}[\tilde\mu, m_1, m_2] -\\
&\frac{1}{8} m_1 m_2 g_1^2 g_2^2\, \mathrm{LF}_{4,1,1,-2}[\tilde\mu, m_1, m_2] + g_1^2 g_2^2\, \mathrm{LF}_{3,2,1,-3}[\tilde\mu, m_2, m_1] +\\
&m_1 m_2 g_1^2 g_2^2\, \mathrm{LF}_{3,2,1,-2}[\tilde\mu, m_2, m_1] - \frac{3}{2}\tilde\mu^4\,\overline{y_d}^{pr}\,\overline{y_d}^{st} y_d^{pt} y_d^{sr}\, \mathrm{LF}_{2,1,1,1,0}[m_{\tilde d}^r, m_{\tilde d}^t, m_{\tilde q}^p, m_{\tilde q}^s] +\\
&\frac{3}{2}\tilde\mu^4\,\overline{y_d}^{pr}\,\overline{y_d}^{st} y_d^{pt} y_d^{sr}\, \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde d}^r, m_{\tilde d}^t, m_{\tilde q}^p, m_{\tilde q}^s] +\\
&\frac{3}{4}\tilde\mu^4\,\overline{y_d}^{pr}\,\overline{y_d}^{st} y_d^{pt} y_d^{sr}\, \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde d}^r, m_{\tilde q}^p, m_{\tilde d}^t, m_{\tilde q}^s] +\\
&\frac{3}{4}\tilde\mu^4\,\overline{y_d}^{pr}\,\overline{y_d}^{st} y_d^{pt} y_d^{sr}\, \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde d}^r, m_{\tilde q}^s, m_{\tilde d}^t, m_{\tilde q}^p] -\\
&\frac{1}{2}\tilde\mu^4\,\overline{y_e}^{pr}\,\overline{y_e}^{st} y_e^{pt} y_e^{sr}\, \mathrm{LF}_{2,1,1,1,0}[m_{\tilde e}^r, m_{\tilde e}^t, m_{\tilde l}^p, m_{\tilde l}^s] +\\
&\frac{1}{2}\tilde\mu^4\,\overline{y_e}^{pr}\,\overline{y_e}^{st} y_e^{pt} y_e^{sr}\, \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde e}^r, m_{\tilde e}^t, m_{\tilde l}^p, m_{\tilde l}^s] +\\
&\frac{1}{4}\tilde\mu^4\,\overline{y_e}^{pr}\,\overline{y_e}^{st} y_e^{pt} y_e^{sr}\, \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde e}^r, m_{\tilde l}^p, m_{\tilde e}^t, m_{\tilde l}^s] +\\
&\frac{1}{4}\tilde\mu^4\,\overline{y_e}^{pr}\,\overline{y_e}^{st} y_e^{pt} y_e^{sr}\, \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde e}^r, m_{\tilde l}^s, m_{\tilde e}^t, m_{\tilde l}^p] +\\
&\frac{1}{8}\tilde\mu^4\,\overline{y_e}^{pr}\,\overline{y_e}^{st} y_e^{pt} y_e^{sr}\, \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde l}^p, m_{\tilde l}^s, m_{\tilde e}^r, m_{\tilde e}^t] +\\
&\frac{3}{8}\tilde\mu^4\,\overline{y_d}^{pr}\,\overline{y_d}^{st} y_d^{pt} y_d^{sr}\, \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde q}^p, m_{\tilde q}^s, m_{\tilde d}^r, m_{\tilde d}^t] -\\
&\frac{9}{4}\tilde\mu^2\,\overline{y_d}^{st} y_d^{pt}\,\overline{a_u}^{pr} a_u^{sr}\, \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde q}^p, m_{\tilde q}^s, m_{\tilde d}^t, m_{\tilde u}^r] + \frac{3}{4}\overline{a_u}^{pr}\,\overline{a_u}^{st} a_u^{pt} a_u^{sr}\\
&\quad \mathrm{LF}_{2,1,1,1,0}[m_{\tilde q}^p, m_{\tilde q}^s, m_{\tilde u}^r, m_{\tilde u}^t] - \frac{3}{4}\overline{a_u}^{pr}\,\overline{a_u}^{st} a_u^{pt} a_u^{sr}\, \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde q}^p, m_{\tilde q}^s, m_{\tilde u}^r, m_{\tilde u}^t] -\\
&\frac{3}{8}\overline{a_u}^{pr}\,\overline{a_u}^{st} a_u^{pt} a_u^{sr}\, \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde q}^p, m_{\tilde u}^r, m_{\tilde q}^s, m_{\tilde u}^t] - \frac{3}{8}\overline{a_u}^{pr}\,\overline{a_u}^{st} a_u^{pt} a_u^{sr}\\
&\quad \left.\mathrm{LF}_{2,2,1,1,-1}[m_{\tilde q}^p, m_{\tilde u}^t, m_{\tilde q}^s, m_{\tilde u}^r] - \frac{3}{4}\overline{a_u}^{pr}\,\overline{a_u}^{st} a_u^{pt} a_u^{sr}\, \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde u}^r, m_{\tilde u}^t, m_{\tilde q}^p, m_{\tilde q}^s]\right)
\end{aligned}
$$